Ideation Phase
Brainstorm & Idea Prioritization

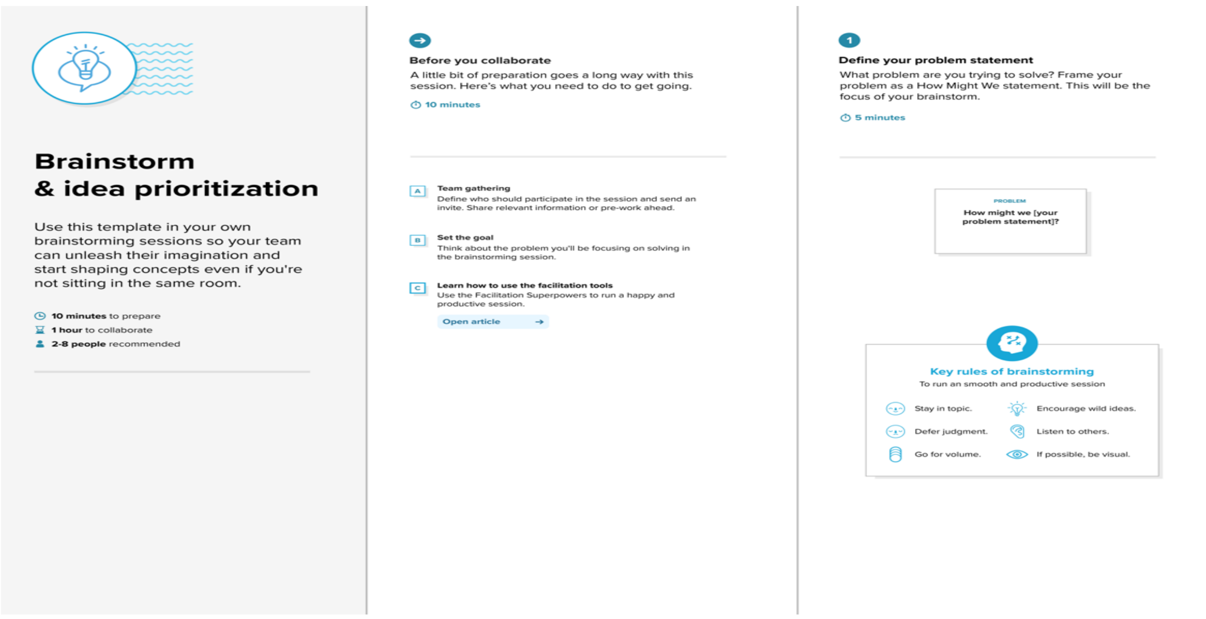
Date	05-07-2026
Team ID	SWTID-2026-8108
Project Name	**Heart Disease Analysis**
Maximum Marks	

Brainstorm & Idea Prioritization :

How might we leverage relational database storage and interactive multi-tab data visualizations to help healthcare researchers instantly identify the co-occurring clinical, socio-demographic, and lifestyle risk factors for heart disease and strokes.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

- **Team Gathering & Pre-work: We set up a team session after importing our core cardiovascular health dataset into our local MySQL instance (schema: data, table: heart_new2). We verified that all 4,416 rows and 18 columns were structurally intact and ready for aggregation.**
- **Collaboration Goal: Our goal was to design an efficient data pipeline to clean our relational data and map out the exact chart configurations needed for an interactive user experience.**
- **Problem Statement: How Might We Statement: How might we leverage relational database storage and interactive multi-tab data visualizations to help healthcare researchers instantly identify the co-occurring clinical, socio-demographic, and lifestyle risk factors for heart disease and strokes?**



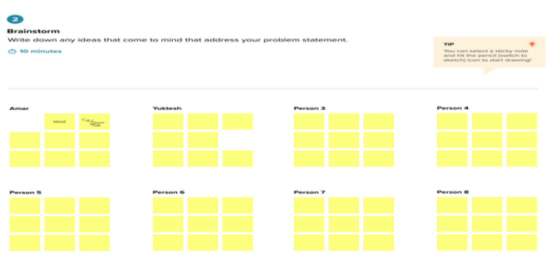
Step-2: Brainstorm, Idea Listing and Grouping

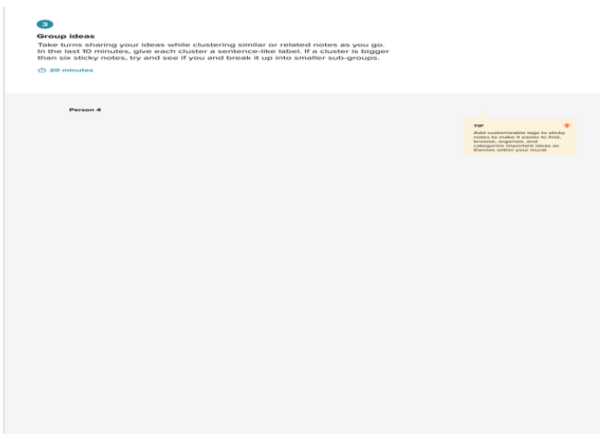
1. Brainstormed Chart Concepts (Idea Listing)

- Clinical Indicators: Bar charts tracking heart disease counts across gender and age distributions; shape plots mapping stroke instances among diabetic patients.
- Socio-Demographic Elements: Pie charts showing racial distributions; packed bubble charts mapping self-reported general health ratings against disease prevalence.
- Lifestyle Footprints: Horizontal stacked bar charts to evaluate how tobacco smoking and heavy alcohol consumption impact cardiovascular diagnoses.
- Comorbidity Matrices: Dense tree maps highlighting dynamic changes in average BMI values across specific age bands and diabetic statuses.

2. Grouped Themes (Final Tableau Storyboard Progression)

- Theme 1: Clinical Risk Factors – Focuses on core medical criteria (Sex, Age, and Diabetes vs. Stroke counts) to serve clinical needs like Dr. Sharma's.
- Theme 2: Socio-Demographic Factors – Maps population traits (Race-wise distribution, General Health Ratings, and Physical Activity status) to serve policy needs like Ramesh's.
- Theme 3: Complex Lifestyle Risks – Visualizes multi-variable correlations (Age and BMI distributions cross-examined by Diabetic profiles) to provide actionable insights for individuals like Anita.





Step-3: Idea Prioritization:

- High Importance / High Feasibility (Core Focus): We prioritized consolidating our data into a multi-tab Tableau Storyboard using an interactive, tiled layout. This gives us the highest clinical value because it lets researchers drill down into complex risk overlaps on a single screen rather than looking at isolated worksheets.

- Data Feasibility Alignment: We chose to manage the dataset using MySQL 8.0 on the InnoDB engine. This ensures that our aggregate calculations—like extracting the average

BMI values across specialized risk cohorts—execute quickly and feed smoothly into our frontend Tableau dashboards.